Pandelis Zembashis

CTO | ML Platform & Infrastructure Leader | Applied AI Systems

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EXECUTIVE SUMMARY

CTO and hands-on engineering leader building applied AI systems, ML-oriented developer infrastructure, and production platforms. At Cosine, lead product and engineering for an AI coding lab built around research-engineering loops: agent orchestration, evaluation-driven product development, secure code execution, memory systems, browser and internet tool use, and enterprise deployment. Previously scaled 11x.ai from 0 to $1M ARR in under six months and have repeatedly built reliable, developer-friendly platforms across Go, TypeScript, React, GraphQL, AWS, Kubernetes, and observability-heavy distributed systems.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Applied ML lab building AI coding agents across CLI, browser, enterprise deployment, desktop, and IDE-oriented surfaces.

- Own product and engineering for Cosine's AI coding platform, aligning model capability, agent behavior, developer experience, reliability, and enterprise deployment into one roadmap.
- Built research-to-product infrastructure around coding agents: browser use, internet research, memory systems, tool orchestration, repository search, and repeatable evaluation loops.
- Architected Cosine CLI as a portable single Go binary with bundled ripgrep and a consistent execution model, reducing friction for internal iteration, external users, and controlled enterprise environments.
- Designed an event-bus agent loop with bidirectional UI events, enabling shared workflows across CLI, desktop, and extension surfaces with clearer observability over state and tool execution.
- Built Kubernetes/gVisor cloud-agent execution for secure code execution, isolated tool use, reliability feedback loops, and private VPC, on-prem, and air-gapped deployment.
- Helped position Cosine as an end-to-end British AI lab and sovereign coding platform, recognised by Tech.eu for benchmark-leading models, 38+ language support, air-gapped deployment, and UK sovereign AI strategy work.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Established onboarding, engineering cadence, and operating rhythms that kept product, engineering, and GTM teams shipping through rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the product API layer by introducing GraphQL into the Ruby on Rails backend, improving iteration across product and engineering.
- Established development cadences and product intake processes that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.

- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

COVERAGE & TECHNICAL COMMUNICATION

Tech.eu on Cosine sovereign AI: Coverage of Cosine as an AI lab and product company building its own models, agents, and full-stack sovereign deployment.
Cosine YC S24 launch: Public writing on building AI software engineering systems and productizing agentic coding workflows.
Papaya engineering profile: Technical leadership profile covering engineering culture, delivery, and cross-functional product work.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

ML Platform & Applied AI: Agent orchestration, research-product loops, evaluations, model-adjacent infrastructure, tool use, memory systems
Platform Engineering & SRE: Go, Kubernetes, gVisor, AWS, observability, logging, tracing, CI/CD, secure execution, incident-aware operations
Product & Engineering Leadership: Technical roadmaps, cross-functional strategy, high-growth hiring, team cadence, enterprise deployment, stakeholder communication
Software Engineering: TypeScript, Node.js, React, Next.js, GraphQL, Ruby on Rails, WebAssembly, CLI tooling, distributed systems